DSAI5101 Statistical Data Mining
Chapter 7: Ranking and Reward Modeling

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Learning Objectives

By the end of this chapter, you should be able to:

- distinguish classification, scoring, and ranking;
- turn pairwise comparisons into a Bradley–Terry (BT) likelihood;
- explain identifiability, comparison graphs, and regularization;
- evaluate rankings on suitable held-out data;
- model response preferences using a conditional BT reward model; and
- learn and evaluate reward scores for reinforcement learning from human feedback (RLHF).

The main bridge from Chapter 6 is the **logistic loss on a difference of scores**.

From Classification to Ranking

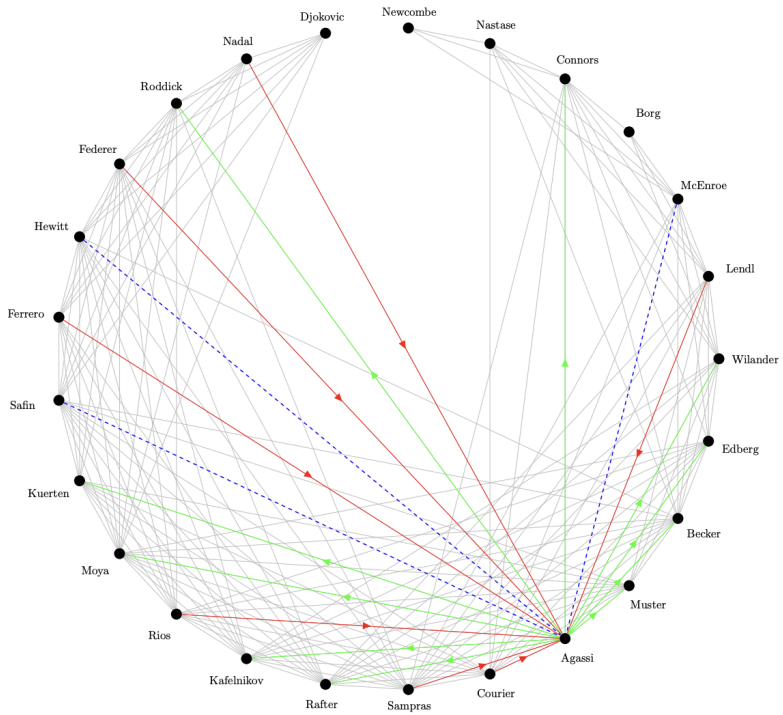
Classification assigns a label to an object. Ranking orders a set of objects according to a criterion.

Task	Example	Output
Classification	Is this email spam?	A class label
Scoring	How relevant is a document?	A numerical score
Ranking	Which documents come first?	An ordered list

A ranking can be obtained by sorting scores. The score need not be a class probability, and its absolute level may have no meaning.

Motivation: Who Is the Better Tennis Player?

Players face different opponents and may never meet directly. Raw win percentages ignore opponent strength.



Agassi

against	Win	Loss
Becker	10	4
Connors	2	0
Courier	5	7
Edberg	6	3
Federer	3	8
Ferrero	2	3
Hewitt	4	4
Kafelnikov	8	4
Kuerten	7	4
Lendl	2	6
McEnroe	2	2
Moya	3	1
Muster	5	4
Nadal	0	2
Rafter	10	5
Rios	1	2
Roddick	5	1
Safin	3	3
Sampras	14	20
Wilander	5	2
Sum	97	85

Comparison network: Bozóki, Csató, and Temesi (2016).

Three Types of Ranking Supervision

Approach	Training signal	Example objective
Pointwise	A label or relevance grade for each item	Predict the grade
Pairwise	Which of two items is preferred	Predict $i \succ j$
Listwise	An ordering or labels for an entire list	Model the ordered list

This chapter focuses on **pairwise comparisons**. In search and response ranking, comparisons are made within the same query or prompt.

A Small Comparison Dataset

Let w_{ij} count the number of times object i defeats or is preferred to object j.

Pair (i, j)	w_{ij}	w_{ji}	Total n_{ij}
(A, B)	6	2	8
(A, C)	3	1	4
(B, C)	5	3	8

- There are 20 comparisons, but only three distinct pairs.
- $n_{ij} = w_{ij} + w_{ji}$; missing pairs have $n_{ij} = 0$.
- We initially assume a clear winner, fixed item strengths, and conditionally independent outcomes.

These are illustrative data for discussing the likelihood.

The Bradley–Terry Model

Give each of n objects a positive strength $s_i > 0$. The BT model specifies

$$p_{ij} = P(i \succ j) = \frac{s_i}{s_i + s_j}.$$

- $s_i = s_j$ gives $p_{ij} = 1/2$.
- $s_i > s_j$ gives $p_{ij} > 1/2$.
- $p_{ji} = 1 - p_{ij}$.
- The odds of i beating j are s_i/s_j.

Rank the objects by their fitted strengths, from largest to smallest.

Bradley and Terry (1952)

BT Is Logistic Regression on Score Differences

Write $u_i = \log s_i$. Then

$$p_{ij} = \frac{e^{u_i}}{e^{u_i} + e^{u_j}} = \sigma(u_i - u_j), \qquad \sigma(t) = \frac{1}{1 + e^{-t}}.$$

Equivalently,

$$\log \frac{p_{ij}}{1 - p_{ij}} = u_i - u_j.$$

A one-unit increase in the score difference multiplies the odds by e.
Sorting u_i gives exactly the same ranking as sorting s_i.

Worked Example: Strengths and Probabilities

Suppose the illustrative strengths are $(s_A, s_B, s_C) = (4, 2, 1)$.

Pair	Probability	Log-odds difference
$A \succ B$	$4/(4+2) = 2/3$	$u_A - u_B = \log 2$
$B \succ C$	$2/(2+1) = 2/3$	$u_B - u_C = \log 2$
$A \succ C$	$4/(4+1) = 4/5$	$u_A - u_C = \log 4$

The ranking is $A \succ B \succ C$, but an upset remains possible. These strengths are assumed for illustration, not fitted to the earlier count table.

Identifiability: Only Relative Strengths Matter

Multiplying every strength by the same $c > 0$ leaves all probabilities unchanged:

$$\frac{cs_i}{cs_i + cs_j} = \frac{s_i}{s_i + s_j}.$$

In log scores, this is the common-shift invariance

$$(u_i + a) - (u_j + a) = u_i - u_j.$$

Fix one normalization, for example $u_n = 0$ or $\sum_i u_i = 0$.

Interpretation

Score differences have meaning. The chosen zero point does not change pairwise probabilities or the ranking.

Likelihood From Win Counts

Under conditional independence, and ignoring constants that do not depend on u, the log-likelihood is

$$\ell(u) = \sum_{i<j} \left[w_{ij} \log p_{ij} + w_{ji} \log(1 - p_{ij}) \right], \qquad p_{ij} = \sigma(u_i - u_j).$$

The maximum likelihood estimator (MLE) of the scores is

$$\hat{u} = \arg \max_{\sum_i u_i = 0} \ell(u).$$

Each observed pair is counted once in the sum. A missing pair contributes no likelihood term. The fitted model can still predict its probability when the objects are linked through other comparisons.

The Logistic Design Matrix

For comparison t, let $b_t = 1$ if i_t beats j_t, and $b_t = 0$ otherwise. With e_i the ith standard basis vector, define

$$d_t = e_{i_t} - e_{j_t}, \qquad P(b_t = 1) = \sigma(d_t^T u).$$

For example, comparing A against C gives $d_t = (1, 0, -1)^T$.

$$-\ell(u) = \sum_t \left[\log(1 + e^{d_t^T u}) - b_t d_t^T u\right].$$

This is Chapter 6's binary logistic loss with an incidence-matrix design. A common item-score intercept cancels in the difference.

Fitting BT Scores

For the negative log-likelihood, the ith gradient component is

$$\frac{\partial(-\ell)}{\partial u_i} = \sum_{j \neq i} \left[n_{ij} \sigma(u_i - u_j) - w_{ij} \right].$$

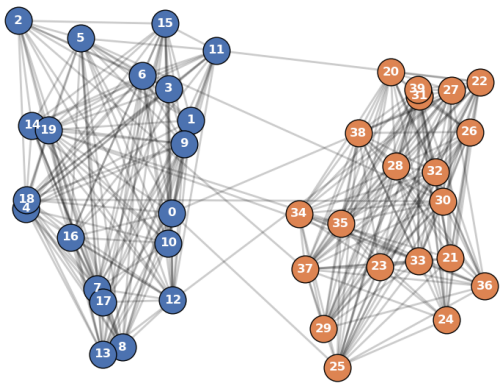
At a finite unpenalized optimum, the model's expected total wins match the observed total wins for each object.

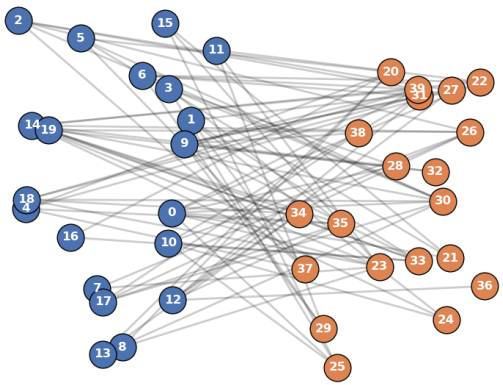
- The loss is convex in u: it is a sum of logistic losses.
- Impose a normalization before optimizing.
- Gradient/Newton methods or specialized BT iterations can be used.
- Numerical convergence does not by itself resolve lack of identifiability or separation.

For specialized iterations: Newman (2023).

Who Is Compared With Whom?

The **undirected comparison graph** has one node per object and an edge whenever $n_{ij} > 0$.

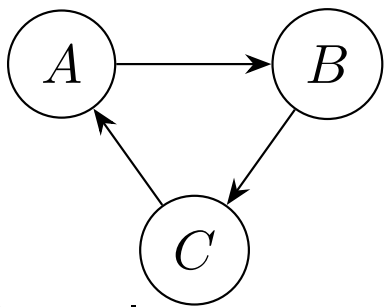




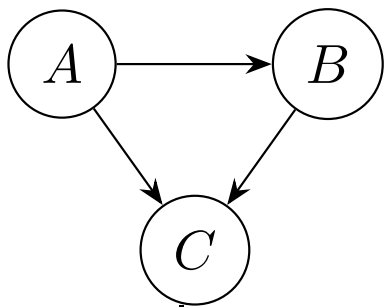
- Connected components have no data linking their relative score levels.
- Sparse links between groups make cross-group comparisons uncertain.
- The graph records comparison opportunities, not who won.

Connected Comparisons Are Not Enough

Draw a directed edge $i \to j$ when $w_{ij} > 0$: **winner to loser**.



Strongly connected



Not strongly connected

For the usual win/loss BT model, a finite, unique unpenalized MLE after normalization exists exactly when the win graph is strongly connected: every node can reach every other node along directed paths.

Separation and Regularization

If A always wins against every opponent it faces, increasing its score relative to the others keeps improving the unpenalized likelihood. A finite MLE may not exist. A practical alternative is

$$\hat{u}_\lambda = \arg \min_{\sum_i u_i = 0} \left\{ -\ell(u) + \frac{\lambda}{2} \|u\|_2^2 \right\}, \qquad \lambda > 0.$$

- The penalty gives finite, unique fitted scores on the constrained space.
- It shrinks extreme estimates, especially for objects with little data.
- For disconnected groups, their relative levels come from the penalty or prior, not from observed cross-group comparisons.

Adding Context and Covariates

A single constant strength may miss age, surface, or home-field effects. For comparison $e = \{i, j\}$, use

$$P(i \succ j \mid e) = \sigma\big(u_i - u_j + (x_{e,i} - x_{e,j})^T v\big)\,.$$

- u_i: an item-specific baseline score.
- $x_{e,i}$: item i's features in this comparison.
- v: shared feature effects.

If $h_{e,i}$ indicates home status, the term $v_h(h_{e,i} - h_{e,j})$ adjusts the log odds for home advantage. Covariate effects require sufficient variation and a suitable identifiable design.

Tennis Case Study

Consider an analysis of ATP tour-level main-draw matches:

- period: 1980–2024;
- 1,803 players and 136,935 matches;
- data source: Jeff Sackmann's tennis_atp repository.

A contextual BT model includes age effects to compare players across career stages.

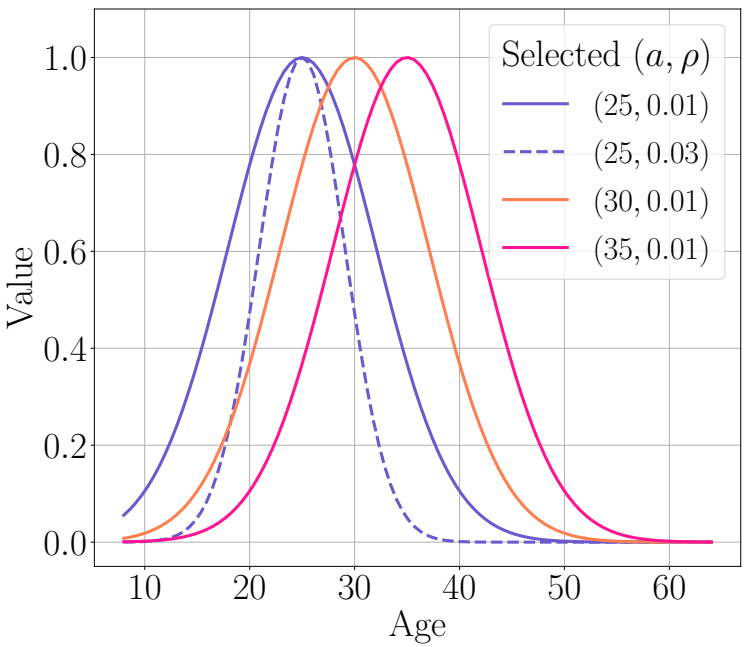
What Is the Ranking Trying to Measure?

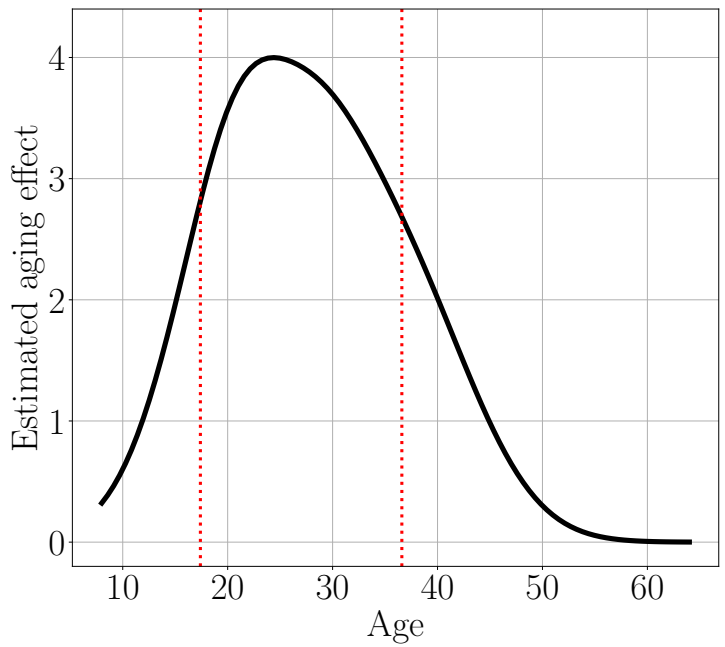
Current form, career achievements, baseline ability, and performance at a given age are different targets. An adjusted ranking answers a model-dependent question.

Age Effects: Features and Fitted Curve

Model a smooth age effect using basis features such as

$$f(t; a, \rho) = \exp\{-\rho(t-a)^2\}.$$





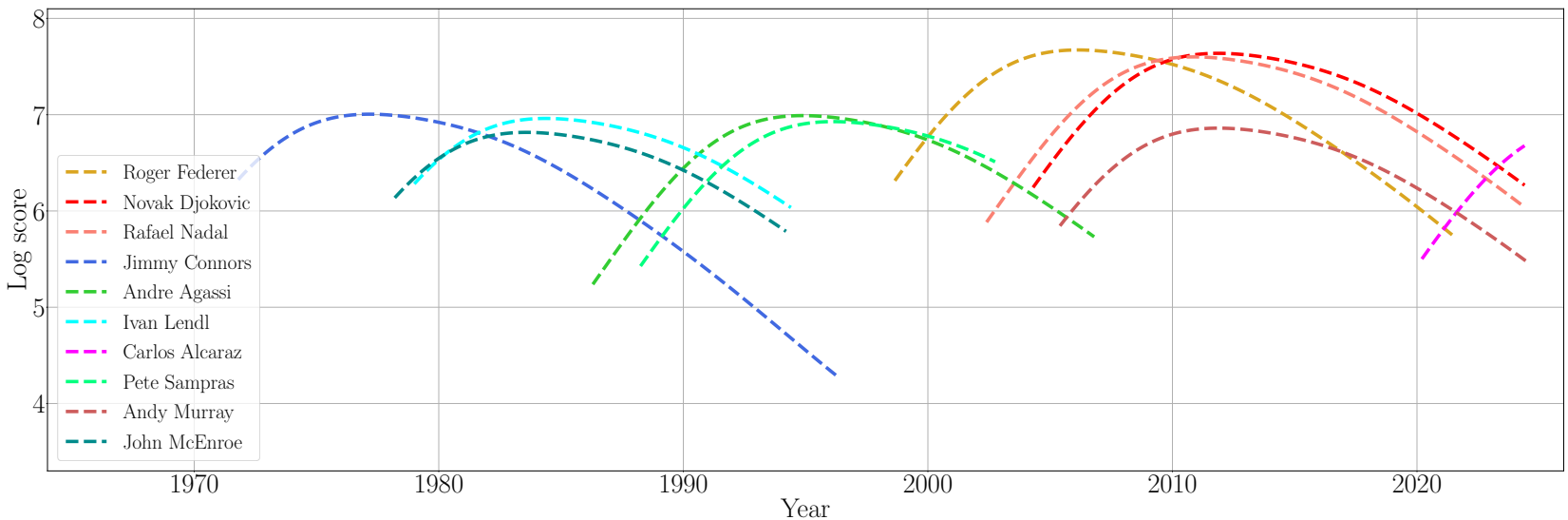
Left: age basis functions. Right: estimated age effect. Red dashed limits contain 99% of observed ages; they are not confidence limits.

A Different Model Can Give a Different Ranking

Player	BT rank	Contextual BT rank
Roger Federer	3	1
Novak Djokovic	1	2
Rafael Nadal	2	3
Jimmy Connors	17	4
Andre Agassi	12	5
Ivan Lendl	5	6
Carlos Alcaraz	4	7
Pete Sampras	7	8
Andy Murray	9	9
John McEnroe	6	10

These model-based ranks illustrate how age adjustment changes the ranking target.

Scores Can Change Across a Career



The plotted trajectories combine player baselines with modeled age effects. A ranking should state its context and time window before we interpret who is “best.”

What a Single-Score BT Model Cannot Capture

BT implies

$$p_{ij} > \tfrac{1}{2}, \quad p_{jk} > \tfrac{1}{2} \quad \Longrightarrow \quad p_{ik} > \tfrac{1}{2}.$$

Thus one fixed scalar score per object cannot represent systematic population preferences like Rock–Paper–Scissors.

- An observed cycle of three match outcomes is still possible under BT because outcomes are random.
- Changing context or heterogeneous judges can produce patterns that a single common score misses.
- Standard BT has no tie outcome; ties need an extension or an explicit data-handling rule.

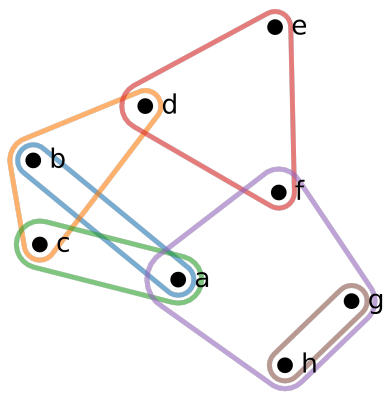
Check whether the model's assumptions match the ranking task.

Optional: From Pairs to Ordered Lists

For a complete ordering $i_1 \succ \cdots \succ i_k$, the Plackett–Luce model gives

$$P(i_1 \succ \cdots \succ i_k) = \prod_{r=1}^{k-1} \frac{s_{i_r}}{\sum_{q=r}^{k} s_{i_q}}.$$

Select the first item from all candidates, then the second from those remaining, and so on. For $k = 2$, this reduces to BT.



Plackett (1975). Pairs extracted from one ranking are dependent, not independent new samples.

Learning a Ranking Function for New Items

Instead of a free score for each known item, learn a shared function $f_\theta(q, i)$ of a query/context q and item features. For a preferred pair $i \succ j$, define

$$\Delta f = f_\theta(q, i) - f_\theta(q, j).$$

Two losses connected to Chapter 6 are

$$\ell_{\log}(\Delta f) = \log(1 + e^{-\Delta f}), \qquad \ell_{\text{hinge}}(\Delta f) = \max(0, 1 - \Delta f).$$

For a linear scorer $f_\theta(q, i) = \theta^T x_{q,i}$, training uses feature differences $x_{q,i} - x_{q,j}$. A shared scorer can generalize to new items whose features are available.

Evaluate Held-Out Pairwise Predictions

For test comparisons (d_t, b_t), predict $\hat{p}_t = \sigma(d_t^T \hat{u})$.

- **Pairwise accuracy**: how often the preferred item gets a higher predicted score.
- **Log loss**: assess the full predicted preference probability:

$$-\frac{1}{T}\sum_{t=1}^{T}\left[b_t \log \hat{p}_t + (1 - b_t)\log(1 - \hat{p}_t)\right].$$

High accuracy alone does not imply calibrated probabilities. State how tied predicted scores are counted, and compare against a simple baseline.

Evaluate a Ranked List: NDCG

With relevance grades, normalized discounted cumulative gain (NDCG) assesses the top of the list. Using nonnegative linear gains $g_r = \mathrm{rel}_r$, define

$$\mathrm{DCG@}k = \sum_{r=1}^{k} \frac{g_r}{\log_2(r+1)}, \qquad \mathrm{NDCG@}k = \frac{\mathrm{DCG@}k}{\mathrm{IDCG@}k}.$$

$\mathrm{IDCG@}k$ is the DCG of the ideal relevance ordering.

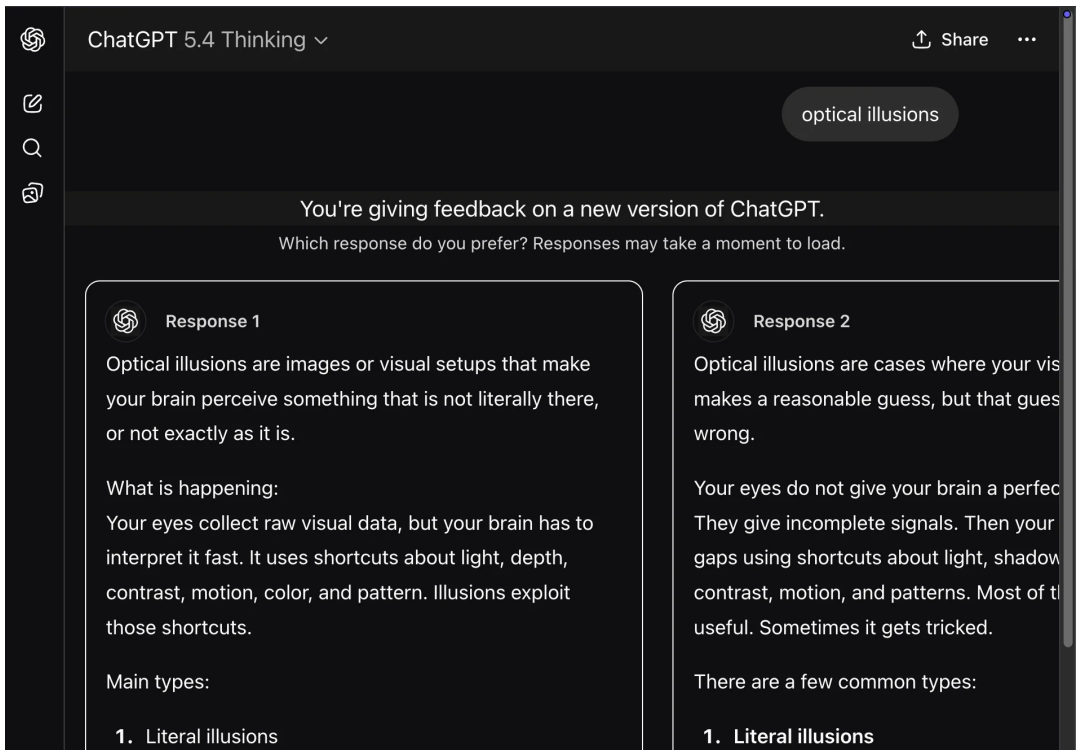
- The discount puts more weight on early positions.
- Compute within each query, then average across queries.
- State the gain convention and how queries with zero ideal gain are handled.

Other implementations use exponential gains $2^{\mathrm{rel}_r} - 1$.

DCG/NDCG were introduced by Järvelin and Kekäläinen (2002). Here we use the common shifted-log discount $\log_2(r+1)$.

From Comparing Players to Comparing Responses

For one prompt, a person can compare two candidate responses. We want a model that learns which response is preferred in that context.



Example of collecting a preference between two responses to the same prompt.

What Does a Preference Example Look Like?

A comparison is (x, y_w, y_l): a prompt, a preferred response, and a less-preferred response.

Prompt x

Explain how ridge regression differs from least squares.

Candidate responses

A: Ridge adds an L_2 penalty to squared error, shrinking coefficients and trading bias for reduced variance.

B: Ridge minimizes exactly the same objective as least squares and always gives the same coefficients.

If the judge prefers A, record $y_w = \mathrm{A}$ and $y_l = \mathrm{B}$. Other tasks may have ambiguous preferences or disagreement between judges.

The Reward Model Is a Conditional BT Model

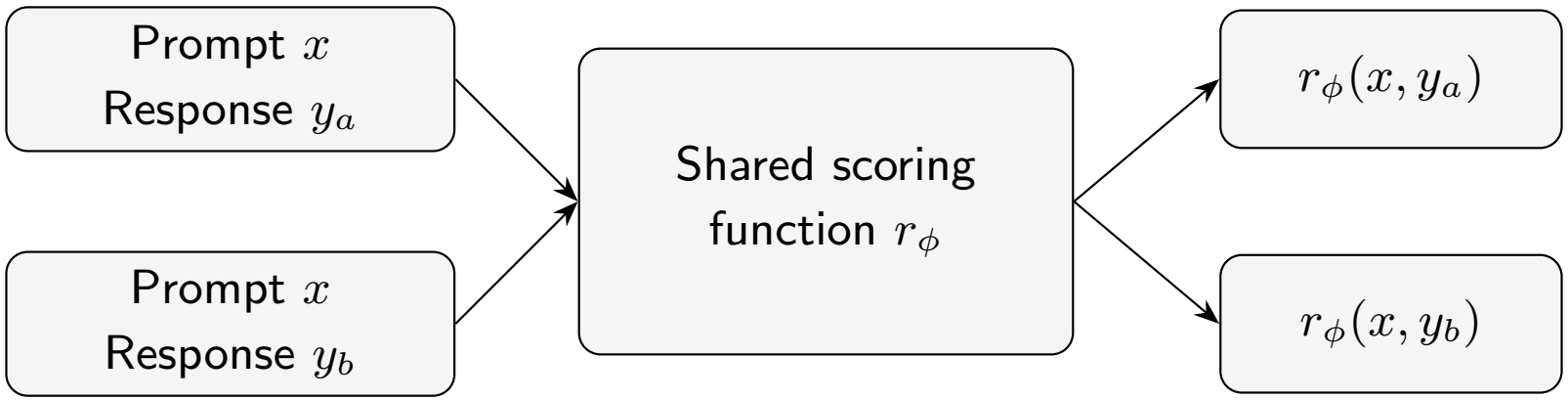
Let $r_\phi(x, y)$ be a scalar score for response y to prompt x. Model the preference probability as

$$P(y_a \succ y_b \mid x) = \frac{\exp(r_\phi(x, y_a))}{\exp(r_\phi(x, y_a)) + \exp(r_\phi(x, y_b))} = \sigma(r_\phi(x, y_a) - r_\phi(x, y_b)) .$$

- Classical BT: one parameter u_i per fixed object.
- Reward model: a shared function of the prompt and response.
- The function can score new responses; its pairwise likelihood has the same logistic form.

One Shared Scorer, Two Candidate Responses

A practical reward model encodes a prompt-response pair and returns one numerical score.



- Both candidates use the same parameters ϕ.
- Subtract the two scores and apply the sigmoid to predict the preference.
- To rank several candidates for one prompt, sort their reward scores.

A scalar neural reward scorer is used in Ouyang et al. (2022).

Training the Reward Model

For a dataset of N preferred/less-preferred pairs, minimize

$$\mathcal{L}_{\mathrm{RM}}(\phi) = -\frac{1}{N}\sum_{t=1}^{N} \log \sigma(r_\phi(x_t, y_{w,t}) - r_\phi(x_t, y_{l,t}))\,.$$

Writing $\Delta r = r_\phi(x, y_w) - r_\phi(x, y_l)$, one pair contributes

$$-\log \sigma(\Delta r) = \log(1 + e^{-\Delta r}).$$

If $\Delta r = \log 2$, the predicted preference probability is $2/3$ and the loss is $\log(3/2) \approx 0.405$. This is precisely the logistic loss from Chapter 6, applied to a pairwise score difference.

Learning From a Ranked List of Responses

For each of M prompts x_m, suppose a judge orders $K_m \geq 2$ responses:

$$y_{m,1} \succ y_{m,2} \succ \cdots \succ y_{m,K_m}.$$

This gives $\binom{K_m}{2}$ preferred pairs. A prompt-balanced pairwise loss is

$$-\frac{1}{M}\sum_{m=1}^{M}\frac{1}{\binom{K_m}{2}}\sum_{a<b}\log\sigma(r_\phi(x_m, y_{m,a}) - r_\phi(x_m, y_{m,b}))\,.$$

- Each of the M prompts receives equal total weight.
- Extracted pairs share responses and are dependent; this is a pairwise training loss, not a joint list likelihood.
- Keep all responses and comparisons from one prompt in the same data split.

What Does a Learned Reward Mean?

Adding any prompt-dependent constant $c(x)$ leaves preference probabilities unchanged:

$$[r_\phi(x, y_a) + c(x)] - [r_\phi(x, y_b) + c(x)] = r_\phi(x, y_a) - r_\phi(x, y_b).$$

- The reward is a fitted proxy for judgments under the training rubric.
- Its absolute zero point is not identified by within-prompt comparisons.
- A higher score does not certify factual correctness or universal agreement.
- Interpret reward differences and rankings within the same prompt.

This is the same shift invariance that appeared in the classical BT model.

Evaluate the Reward Model as a Ranker

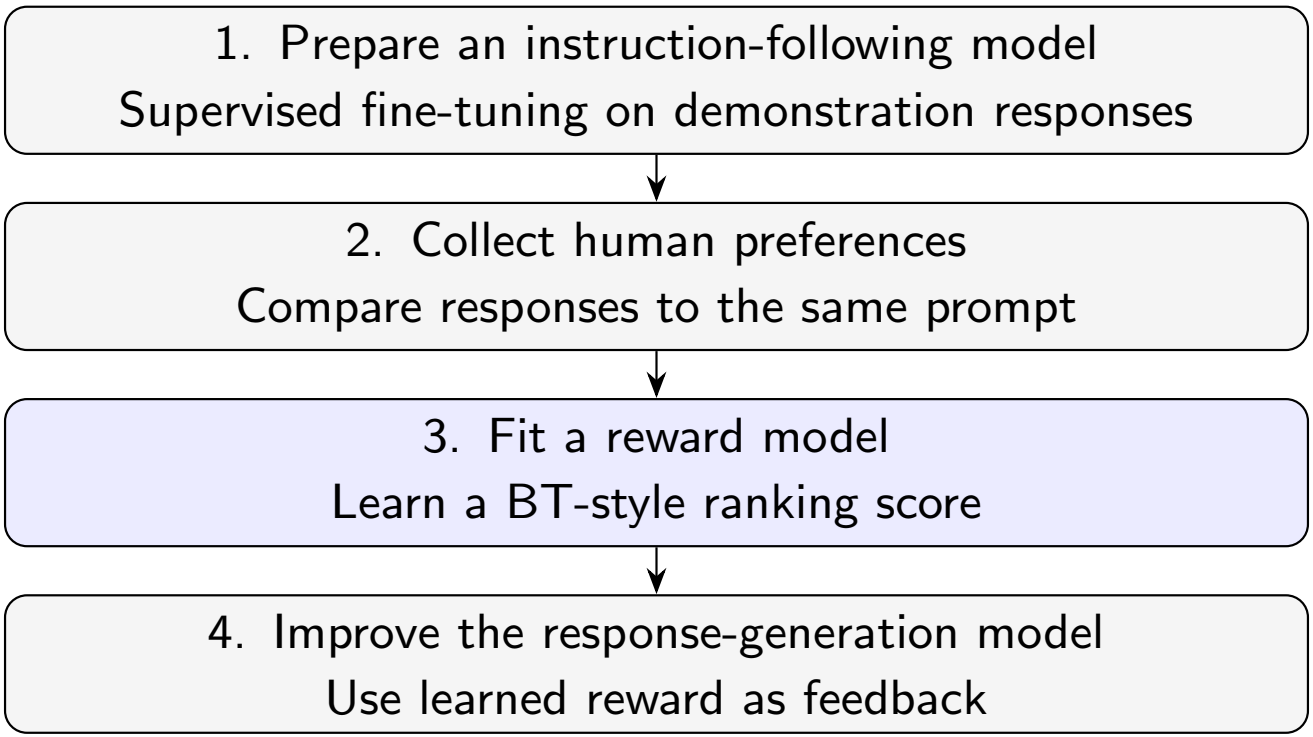
On held-out prompts and response pairs, assess:

- **Preference accuracy**: does the preferred response get a higher score?
- **Log loss and calibration**: do predicted preference probabilities match observed choices?
- **List quality**: when relevance or quality grades are available, assess the ranked list with a stated metric such as NDCG.
- **Group performance**: inspect results across topics, judges, and user groups.

Compare against a simple baseline. Use independent judgments, and report uncertainty and judge disagreement.

Brief Context: The Classical RLHF Pipeline

Reinforcement learning from human feedback (RLHF) uses human judgments to guide a response-generation model.



Ouyang et al. (2022). The statistical focus here is the preference data and reward model.

Brief Context: How Reward Scores Are Used

The same learned scorer can support two different uses:

Rank or select existing candidates

Generate several responses to one prompt, score them with $r_\phi(x, y)$, and select a highly ranked candidate.

Guide improvement of the generator

Use reward scores as feedback to train the generator to produce more preferred responses. This is the reinforcement-learning stage in the classical RLHF pipeline.

In both cases, the reward is a proxy for human judgments. Check the quality of selected or newly generated responses using independent evaluation.

Stiennon et al. (2020); Ouyang et al. (2022).

Chapter Summary

- Ranking orders objects; pairwise data tell us which object is preferred.
- BT uses $P(i \succ j) = \sigma(u_i - u_j)$ and is logistic regression on score differences.
- Normalization, comparison structure, and separation matter when estimating scores.
- Contextual effects change the question answered by a ranking.
- A reward model learns prompt-conditioned response scores using the same BT pairwise loss.
- Reward differences predict preferences; their absolute zero point is not identified.
- Evaluate on held-out prompts, checking ranking quality, probability calibration, and judge disagreement.

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These papers connect human preference data, pairwise ranking losses, and learned reward scores.